

End-to-End Data Management Pipeline for a Recommendation System

RecoMart Data Platform Team

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1. Problem Statement

RecoMart, an e-commerce startup, wants to build a data-driven recommendation engine to improve customer engagement and cross-selling conversion. User behavior is captured across multiple heterogeneous sources -- web/mobile clickstream logs, transactional purchase history, product catalog metadata, and external sentiment/popularity signals -- and must be continuously ingested, validated, curated into features, and used to train/update a personalized recommendation model.

2. Objectives

- Ingest at least two heterogeneous data types (CSV-based interaction logs, REST API product catalog) on an automated, retryable schedule.
- Store raw data in a partitioned local data lake (source / type / date).
- Profile and validate data quality (missing values, duplicates, schema, range checks) with an automated report.
- Clean, encode, and normalize data; conduct EDA on interaction distributions, item popularity, and sparsity.
- Engineer recommendation-ready features (activity frequency, average ratings, item co-occurrence) into a structured warehouse.
- Stand up a lightweight, versioned feature store serving both training and inference.
- Version raw/processed data and track transformation lineage.
- Train and evaluate a collaborative-filtering recommendation model (Precision@K, Recall@K, NDCG@K), tracked via MLflow.
- Orchestrate the full pipeline end-to-end with automated retries and logging.

3. Methodology / Pipeline

The pipeline is implemented as 6 modular stages, each independently runnable and orchestrated end-to-end with Prefect:

- 1) Ingestion -> 2) Raw Storage -> 3) Validation -> 4) Preparation (Cleaning + EDA)
- > 5) Transformation (Feature Engineering into SQLite warehouse)
- > 6) Feature Store retrieval -> 7) Model Training & Evaluation (MLflow).

Data sources: (a) clickstream events and (b) transaction/purchase records are simulated per run to mimic a live Kafka/OLTP source system; (c) product catalog is fetched live from the FakeStore REST API (<https://fakestoreapi.com/products>); (d) product sentiment/popularity scores are retrieved via a retry-capable scoring-service client (simulated locally, same integration pattern as a real API).

4. Implementation Details

- Ingestion: `src/ingestion/*.py` - tenacity-based exponential-backoff retries, rotating file logs (`logs/ingestion.log`), partitioned raw storage.
- Validation: `src/validation/validate_data.py` - pandas-based schema/null/duplicate/range checks -> JSON + PDF quality report.
- Preparation: `src/preparation/clean_and_prepare.py` - dedup, missing-value handling, LabelEncoder/MinMaxScaler, EDA plots.
- Transformation: `src/transformation/feature_engineering.py` + `schema.sql` - user/item/co-occurrence features loaded

into SQLite warehouse (warehouse/recomart.db).

- Feature Store: src/feature_store/feature_store.py + feature_registry.yaml - versioned, declarative feature metadata with offline/online retrieval.
- Modeling: src/models/train_model.py - Truncated SVD collaborative filtering over a weighted implicit+explicit interaction matrix; src/models/inference.py - top-N recommendation interface.
- Orchestration: src/orchestration/pipeline_flow.py - Prefect @flow/@task DAG with retries and structured logging.
- Versioning: docs/VERSIONING.md - Git + DVC workflow for data/model lineage.

5. Results and Output

Warehouse table row counts (latest pipeline run):

- dim_users: 495
- dim_items: 20
- fact_interactions: 2000
- fact_transactions: 400
- item_cooccurrence: 190

Model evaluation metrics (MLflow-tracked, latest run):

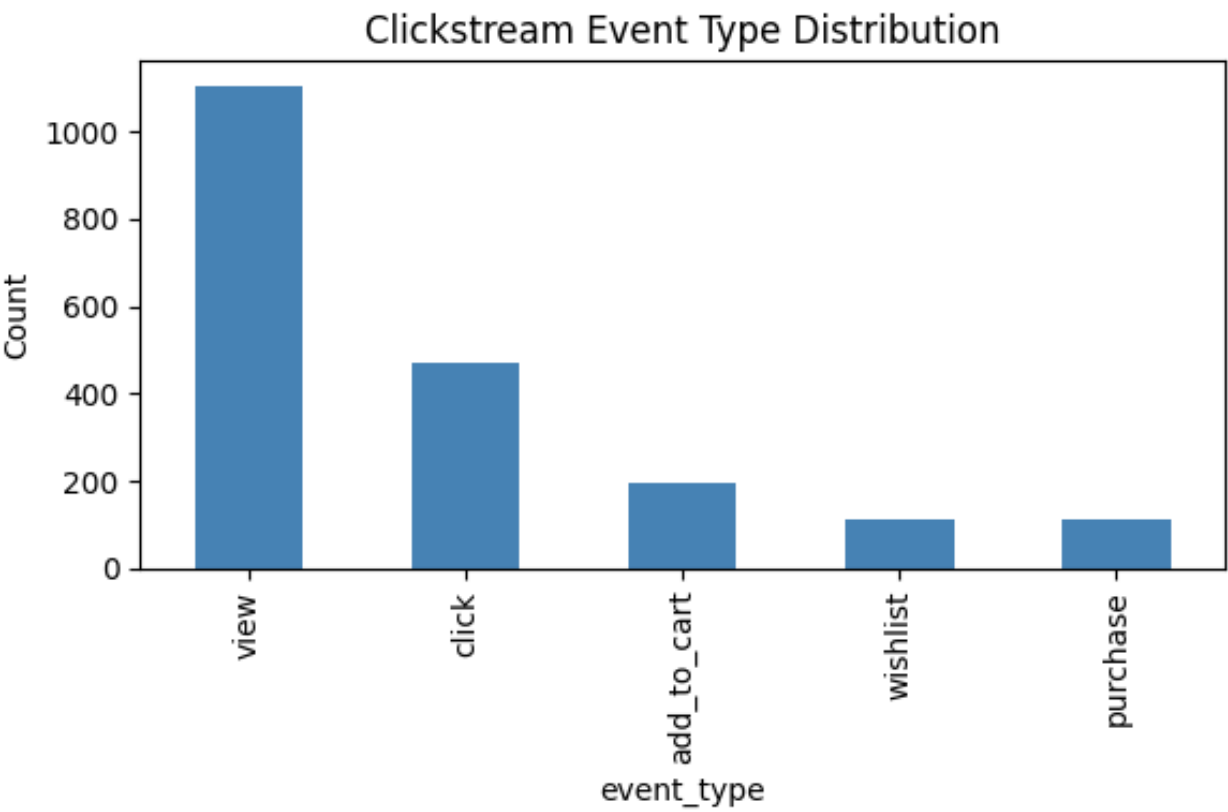
- explained_variance_ratio: 0.9751
- ndcg_at_k: 0.3220
- precision_at_k: 0.0675
- recall_at_k: 0.5508

Data quality validation summary:

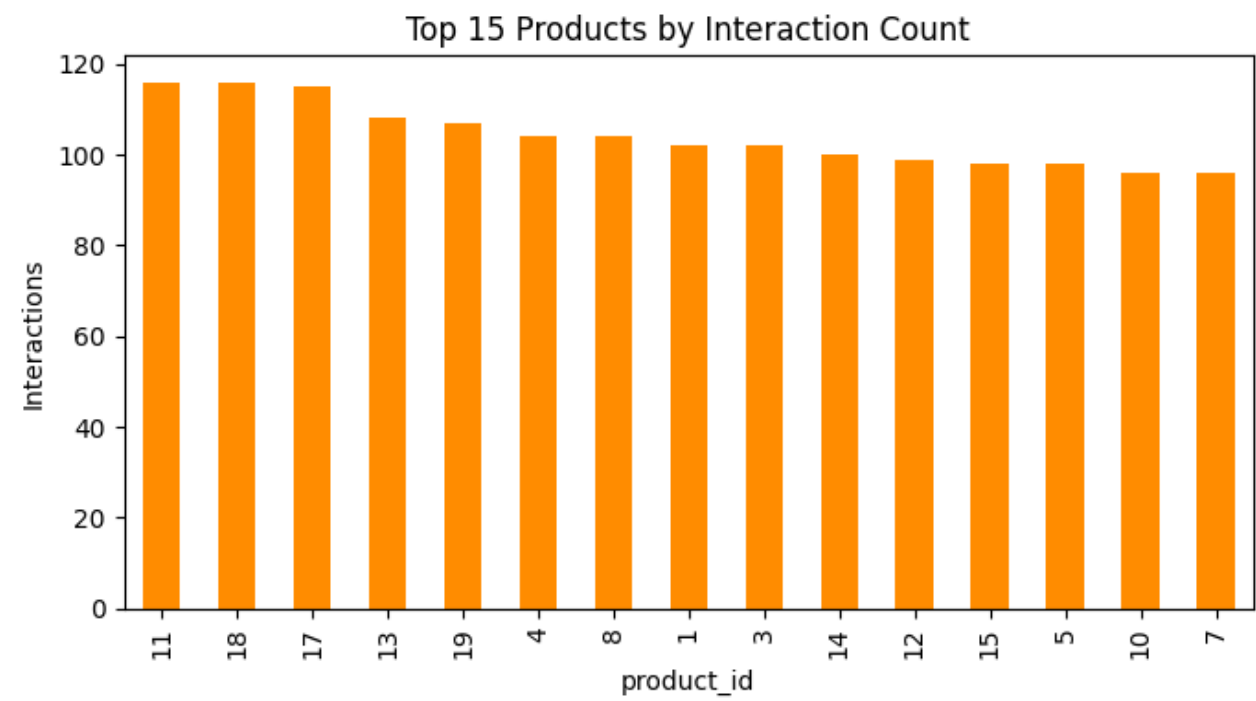
- clickstream: status=OK, rows=2000, issues=0
- transactions: status=OK, rows=400, issues=0
- products: status=OK, rows=20, issues=0
- sentiment: status=OK, rows=20, issues=0

EDA Summary Plots:

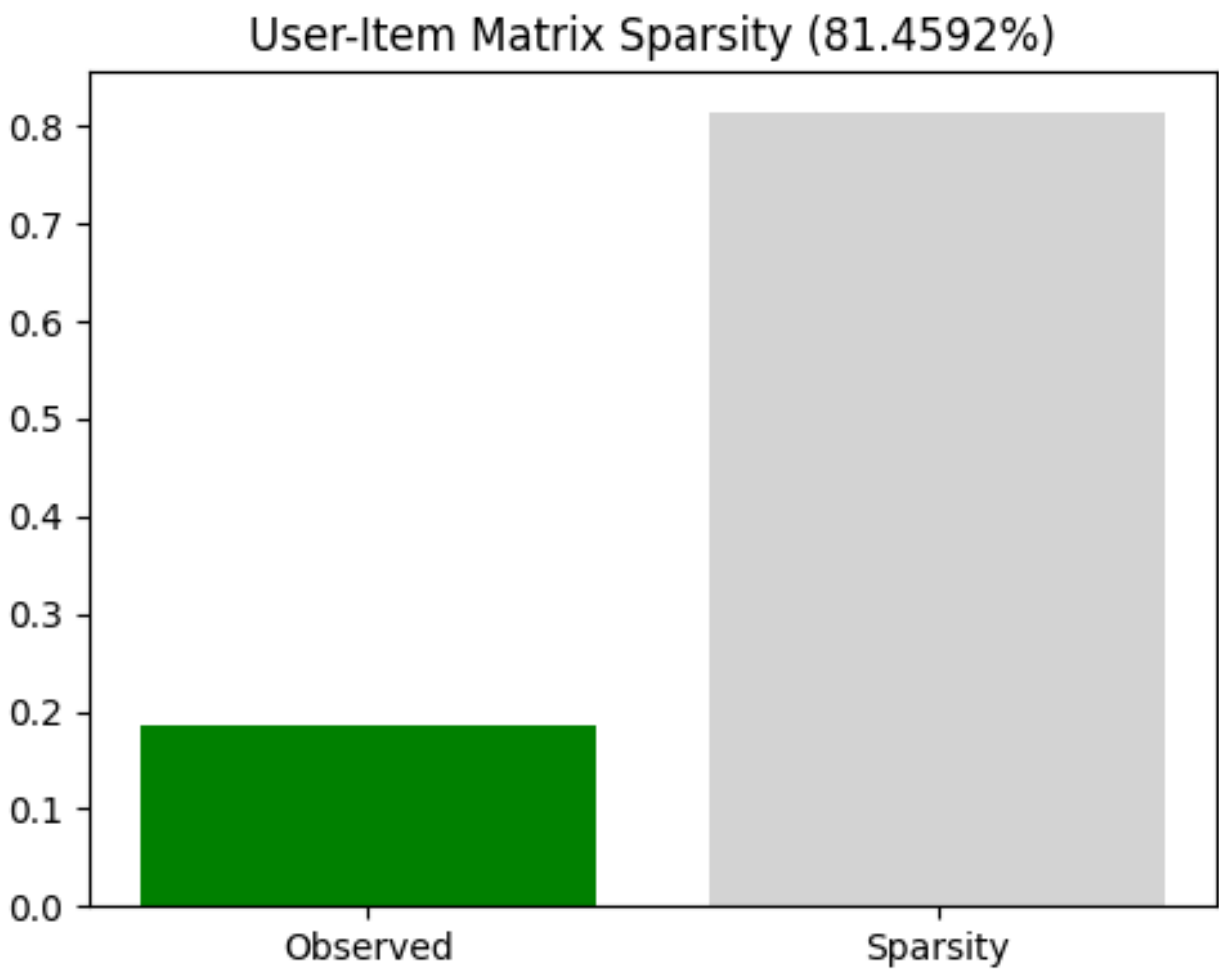
Event Type Distribution



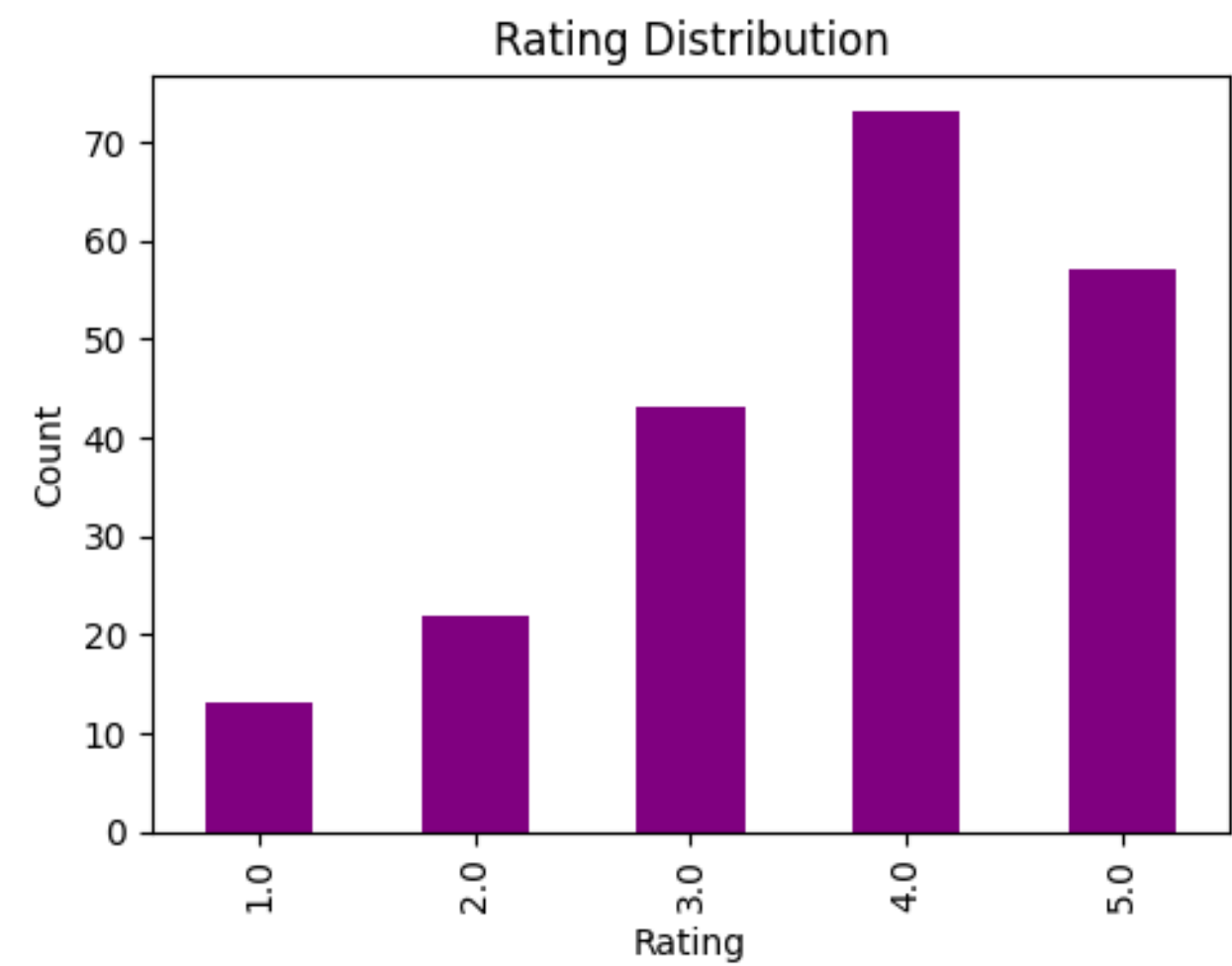
Item Popularity



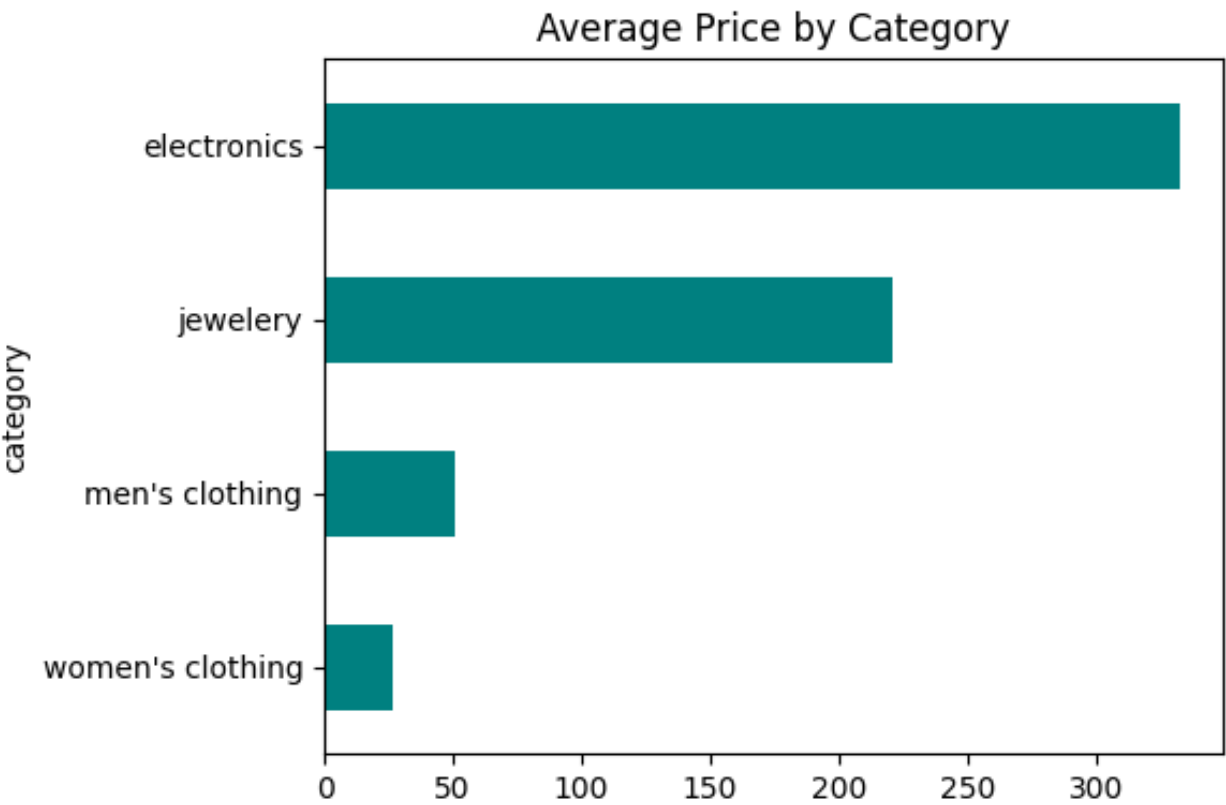
Sparsity



Rating Distribution



Avg Price By Category



6. Conclusion and Future Scope

This POC demonstrates a fully modular, end-to-end data management pipeline covering ingestion, validation, preparation, feature engineering, a versioned feature store, model training/evaluation, and orchestration -- aligned with modern data-stack and MLOps practices. The trained SVD collaborative-filtering model achieves measurable Precision@10 / Recall@10 / NDCG@10 on held-out interactions, and the pipeline is fully re-runnable end-to-end via a single Prefect flow.

Future scope:

- Replace simulated clickstream/transaction sources with a real Kafka topic / OLTP CDC feed.
- Add content-based filtering using product text embeddings (title/description) blended with the collaborative model.
- Migrate raw storage to a real cloud data lake (S3/ADLS) with lifecycle policies.
- Adopt Feast for a production-grade online feature store with a low-latency serving layer.
- Schedule the Prefect flow on a deployment/work-pool for true periodic automation with alerting on failures.
- A/B test the recommendation model in a live RecoMart storefront and track online CTR/conversion uplift.

7. Submission Links

Video walkthrough (Google Drive): <PASTE GOOGLE DRIVE VIDEO WALKTHROUGH LINK HERE>

Project deliverables .zip (Google Drive): <PASTE GOOGLE DRIVE PROJECT ZIP LINK HERE>